Late Breaking Results: Decentralized Voting-Based Attestation for IoT Devices

Mohamed Alsharkawy*, Eren Sönmez*, Jeferson González-Gómez*†, Hassan Nassar*, Jörg Henkel*
*Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany
†Instituto Tecnológico de Costa Rica (TEC), Cartago, Costa Rica
Corresponding Author: mohamed.alsharkawy@kit.edu

***Abstract*—Remote Attestation (RA) has become a valuable security service for Internet of Things (IoT) devices, as the security of these devices is often not prioritized during the manufacturing process. However, traditional RA schemes suffer from a single point of failure because they rely on a trusted verifier. To address this issue, we propose a voting-based blockchain attestation protocol that provides a reliable solution by eliminating the single point of failure through distributed verification across all nodes. In addition, it offers a traceable and immutable public history of the attestation results, which can be verified by external auditors at any time. Finally, we verify our proposed protocol on three NVIDIA Jetson embedded devices hosting up to 15 attestation nodes.**

***Index Terms*—Remote Attestation, Blockchain, Decentralized Systems, Embedded Devices, IoT**

I. Introduction

In recent years, the number of Internet of Things (IoT) devices has grown significantly, with applications in areas such as smart homes, smart cities, and smart factories. While connecting these devices to the Internet provides numerous benefits, it also increases their exposure to cyber threats, making security more complex. Due to cost, size, and power limitations, manufacturers often do not prioritize security, making it difficult for these devices to prevent attacks [1].

Remote Attestation (RA) offers a practical security solution by assessing whether a device has been compromised. A practical example of RA can be found in industrial IoT environments, such as smart factories. In this scenario, external authority acts as the verifier who needs to ensure that connected devices, including sensors, robotic arms, and controllers (known as provers), are running trusted firmware and remain free from malware. The verifier initiates the process by sending a challenge to the prover, requesting evidence of its integrity. The prover responds by generating a cryptographic report containing a proof of integrity e.g., measurements of its firmware, configuration, or runtime state [2], which is securely signed using a Trusted Platform Module (TPM) or similar secure hardware. The verifier then compares the reported measurements with known trusted values. If they match, the device is confirmed secure.

Although RA is widely accepted, traditional approaches face privacy and security concerns, such as the single point of failure. Since the verifier is assumed to be trusted, it could be compromised or malicious, leading to the failure of the attestation process [3]. Although the attested-by-neighbor [4] scheme allows a node to be verified by multiple neighbors and aggregates the results at a cluster head for the final attestation decision, it does not provide a fully decentralized solution. The cluster head itself could be malicious, or the neighboring nodes could be compromised. Furthermore, the scheme lacks transparency and immutability, as the verification results from neighboring nodes are only accessible to the cluster head and are not made publicly available. To address these limitations, we propose a functional proof-of-concept of a lightweight decentralized attestation protocol for embedded devices.

This work was partially funded by the German Federal Ministry of Education and Research (BMBF) through grant 01IS23066 – as part of the Software Campus Projects "HE-Trust" and "VERITAS".

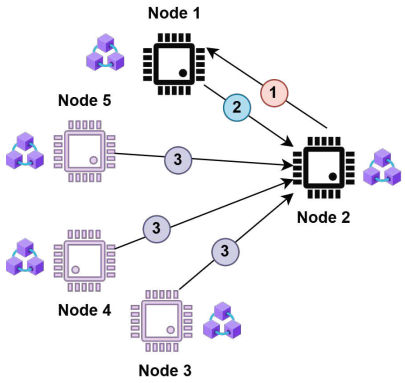

Fig. 1: Attestation process over blockchain: Two random nodes perform the attestation. ① The initiator sends a challenge to the prover. ② The prover responds with proof on integrity. ③ each node verifies the response from the distributed ledger and submit a transaction with a decision.

In this protocol, all eligible participant nodes act as verifiers instead of relying on a single trusted verifier that could be compromised. We are the first to utilize blockchain to facilitate a secure voting process among the verifier nodes for attestation. Blockchain has proven to be effective in distributed electronic voting systems for national elections, as demonstrated in [5], [6]. It permanently records the verification results of nodes in a distributed ledger, ensuring the immutability and transparency of the votes. This allows external auditors to conduct majority voting and independently verify the attestation results.

II. Decentralized voting-based attestation for IoT

Our approach eliminates the need for a trusted verifier while also addressing the single point of failure. In our technique, all nodes participate in the verification process, ensuring system integrity even if some nodes are malicious or compromised.

The assumption that more than one node can act as verifiers aligns with attested by neighbor approach [4].

The proposed solution utilizes blockchain as a voting technique to count the verifiers' votes, where the verification process is determined by the majority voting of all nodes in the blockchain. In our technique, all nodes participate in the verification process, ensuring system integrity even if some nodes are malicious or compromised. Attestation will only fail if an attacker successfully compromises 51% or more of the nodes, as this would disrupt the final voting calculation performed by the external auditors.

Figure 1 illustrates how the attestation process is conducted on the blockchain. In ①, the protocol selects two random nodes: the ***initiator***, who generates a random challenge and sends a verification request along with the challenge to the other node, known as the ***prover***, which needs to be attested. This request is represented as a transaction in the distributed ledger. In ②, once this transaction is mined into the blockchain, the prover automatically sends a transaction back to the initiator with its proof of integrity linked to the corresponding challenge. After this second transaction is mined, the voting process begins in ③, where each other node in the network acts as a **verifier** i.e., all other nodes check the prover's response against the initiator's challenge and submit transactions with their decisions. These votes are then mined and permanently stored on the blockchain, enabling any external audit to independently check the attestation results by counting the votes.

III. Experimental Evaluation

a) Experimental setup: To verify the proposed protocol, a Python-based blockchain with a proof-of-work consensus algorithm was developed, running on three NVIDIA Jetson Nano boards, as shown in Figure 2. Each board hosts up to five independent nodes, resulting in a total of 15 nodes on the blockchain. The block miners are randomly selected by the protocol.

Fig. 2: NVIDIA Jetson embedded devices used in the setup.

b) Security Analysis: To evaluate the security of our protocol we present two possible scenarios. In the first scenario, the prover node is compromised. Here, the protocol effectively detects such a compromise, as verifier nodes are expected to identify the compromised status of the prover by examining the distributed ledger, where the prover's response, attestation request, and challenge are permanently stored. In addition, the decisions of all the verifier nodes are also recorded in the distributed ledger. This immutable and transparent record allows external auditors to independently verify the correctness of the attestation process at any time. In the second scenario N verifier nodes are compromised, where N can reach up to 49% of the total nodes. These compromised nodes may intentionally make incorrect decisions to manipulate the attestation process. However, since all votes are permanently publicly recorded in the distributed ledger, external auditors can independently perform majority voting over the recorded votes. This allows identification of malicious or malfunctioning verifier nodes that made erroneous decisions and acts as a deterrent to such actions, as these nodes will ultimately expose themselves.

c) Performance Analysis: We analyze the timing of the attestation process, which consists of four main steps. The first step involves the initiator sending an attestation request with a challenge. The second step occurs when the prover retrieves the challenge from the blockchain and responds with a proof of integrity. In the third step, each node validates the prover's response and records its vote. All three of these steps have $O(1)$ complexity and are independent of the size of the network. However, the vote counting step has a complexity of $O(n)$, where n is the number of voting nodes. We evaluate this step on a NVIDIA Jetson Nano board with varying numbers of voting nodes as shown in Figure 3. The figure shows that even with embedded devices, the latency remains minimal, with only 40 ms required when 15 nodes participate.

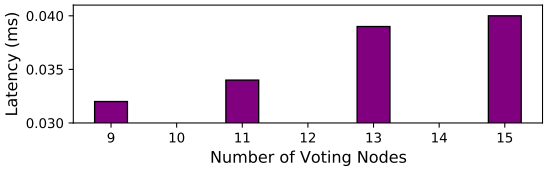

Fig. 3: Vote counting latency for a challenge-response transaction.

IV. Conclusion

In this paper, we propose a proof of concept of a decentralized voting-based attestation scheme for IoT devices utilizing blockchain technology, which addresses the single point of failure challenge faced by traditional approaches. Additionally, the protocol offers an immutable and traceable history of attestation results, allowing any participant to check and trace the process. From our experimental evaluation on three NVIDIA Jetson Nano boards, it is clear that a decentralized voting-based attestation protocol is feasible on real hardware with very low verification latency and constant time complexity on the prover's side, making our proposed solution efficient and reliable for IoT security.

References

[1] W. H. Hassan *et al.*, "Current research on internet of things (iot) security: A survey", *Computer networks*, vol. 148, 2019.

[2] J. Gonzalez-Gomez *et al.*, "Lightfat: Mitigating control-flow explosion via lightweight pmu-based control-flow attestation", in *HOST*, 2024.

[3] R. V. Steiner *et al.*, "Attestation in wireless sensor networks: A survey", *ACM Computing Surveys (CSUR)*, vol. 49, no. 3, 2016.

[4] B. Kuang *et al.*, "A survey of remote attestation in internet of things: Attacks, countermeasures, and prospects", *Computers & Security*, 2022.

[5] F. T. Hjálmarsson *et al.*, "Blockchain-based e-voting system", in *International Conference on Cloud Computing*, 2018.

[6] J. Huang *et al.*, "The application of the blockchain technology in voting systems: A review", *ACM Computing Surveys*, vol. 54, no. 3, 2021.